



Understanding DNS – Part 2

This lesson continues our discussion on the domain name system.

In this lesson, you will get an insight into the complete *DNS query lookup* process and understand the role of different servers in the *DNS* infrastructure.

Lets' begin.

When the user hits enter after typing in the domain name into their browser, the browser sends a request to the *DNS Recursive nameserver*, also known as the *DNS Resolver*.

The role of *DNS Resolver* is to receive the client request and forward it to the *Root nameserver* to get the address of the *Top-Level domain nameserver*.

The *DNS Recursive nameserver* is generally managed by our *ISP* (Internet service provider). The whole *DNS* system is a distributed system setup in large data centers managed by internet service providers.

These data centers contain clusters of servers optimized to process *DNS* queries in minimal time in milliseconds.

Once the *DNS Resolver* forwards the request to the *Root nameserver*, the *Root nameserver* returns the address of the *Top-Level domain nameserver* in response. As an example, the top-level domain for *amazon.com* is *.com*.



Once the *DNS Resolver* receives the address of the *top-level domain nameserver*, it sends a request to it to fetch the details of the domain name.



Top Level domain nameservers hold the data for domains using their top-level domains.

> For instance, the *.com* top-level domain nameserver will contain information on domains using *.com*. Similarly, a *.edu* top-level domain nameserver will hold information on domains using *.edu*.

Since our domain is *amazon.com*, the *DNS Resolver* will route the request to the *.com* top-level domain name server.

Once the top-level domain name server receives the request from the *Resolver*, it returns the *IP address* of the *amazon.com* domain name server.

amazon.com domain nameserver is the last server in the *DNS query lookup* process. This nameserver is responsible for the *amazon.com* domain and is also known as the *Authoritative nameserver*. The owner of the domain name owns this nameserver.

Then, *DNS Resolver* fires a query to the *Authoritative nameserver*, and it returns the *IP address* of the *amazon.com* website to the *DNS Resolver*.

DNS Resolver caches the data and forwards it to the client.

On receiving the response, the browser sends a request to the *amazon.com* website's *IP address* to fetch data from their servers.

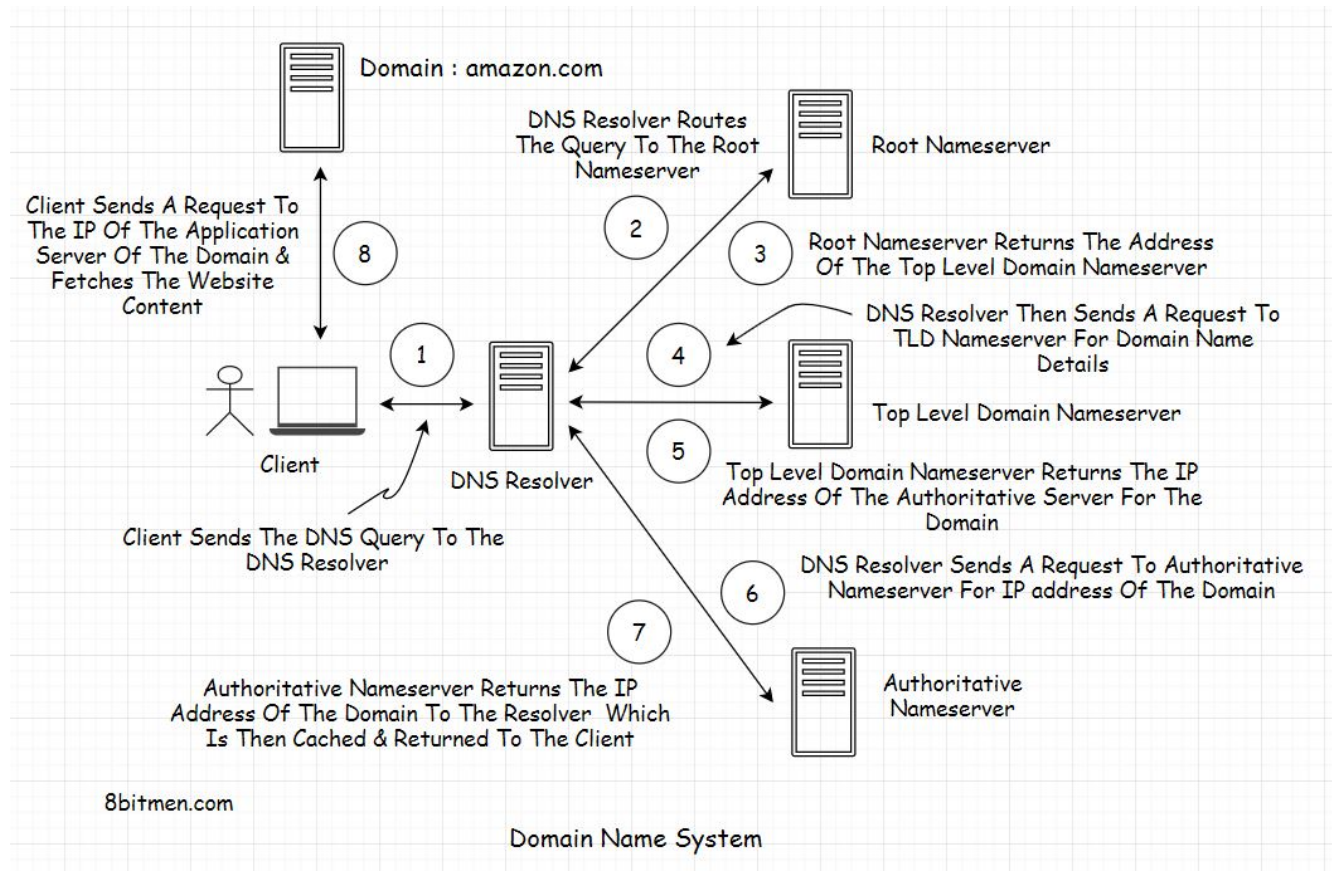
Often all this *DNS* information is cached, and the *DNS* servers don't have to do so much rerouting every time a client requests an *IP* of a certain website.

The *DNS* information of websites that we visit also gets cached in our local machines, that is, our browsing devices with a *TTL* (Time to live). ?

All modern browsers do this automatically to cut down the *DNS* query lookup time when revisiting a website. Tt



This is how the entire *DNS query lookup* process works.



In the next lesson, let's get an insight into *DNS load balancing*.

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